29 A radioactive sample consists of two nuclides.

One has a half-life of 48 h and decays by emitting alpha particles, while the other has a half-life of 36 h and emits beta particles.

The initial count rate is 352 min^{-1}. When a sheet of paper is placed between the source and the detector, the count rate becomes 256 min^{-1}. The background count rate is 16 min^{-1}.

What will the count rate be after 6 days, without the paper present?

A 27 min^{-1} **B** 36 min^{-1} **C** 43 min^{-1} **D** 52 min^{-1}